SOFTWARE ENGINEERING LAB (EXPERIMENT 7)

Assignment 2: Testing, CI/CD Pipelines & Agile Collaboration Report

Student Name: Sooraj Jose George	**Roll Number:** 60
GitHub Username: SoorajGeorge123	**Role / Module:** Face Detection Module
Project Title: SAFE-VISION: AI-Powered Media Moderation & Quality Assurance System	**Subject Code:** CS331
Collaboration Repo: RDSOURAV05/software_engineering_lab_experiment_7_assignment	**Feature Branch:** feature-sooraj

Part 1: GitHub Issues & Branching

In this step, an issue was created in the collaboration repository **RDSOURAV05/software_engineering_lab_experiment_7_assignment** to define requirements and track progress for the Face Detection Module. The issue titled **'Implement testing for Face Detection Module.'** was assigned to **SoorajGeorge123**. A dedicated feature branch named **feature-sooraj** was created locally using `git checkout -b feature-sooraj` to isolate module development and unit tests.



Figure 1 (Screenshot 1 Required): Assigned Issue #3 showing Title, Description, and Assignee (SoorajGeorge123).

Part 2: Unit Testing Suite & CI/CD Pipeline Implementation

The Face Detection Module logic was implemented in `face_detection.py` featuring format validation, empty/missing checks, resolution validation, and detection logic. A comprehensive PyTest suite was authored in `test_sooraj.py` covering 5 test cases including Boundary Value Analysis (BVA) for 0x0 resolution and Equivalence Partitioning (EP) for corrupted inputs and unsupported formats. The automated CI/CD pipeline was configured under `.github/workflows/ci.yml` to trigger on both push and pull request events.

Sequential Git Commits Performed:

1. **Commit 1:** "Added unit tests for Face Detection Module" (test suite & face_detection module)
2. **Commit 2:** "Updated README for feature-sooraj" (required documentation update)
3. **Merge Commit:** "Merge origin/main and resolve conflicts for feature-sooraj" (synced latest team updates)

Part 3: Pull Requests, Peer Review & GitHub Actions (CI/CD)

A Pull Request (PR #9) titled **"Add tests and CI for Sooraj."** was created from `feature-sooraj` targeting `main`. Code review was requested from peer collaborator **RDSOURAV05**, who approved the pull request. All GitHub Actions automated CI tests executed successfully with zero merge conflicts.

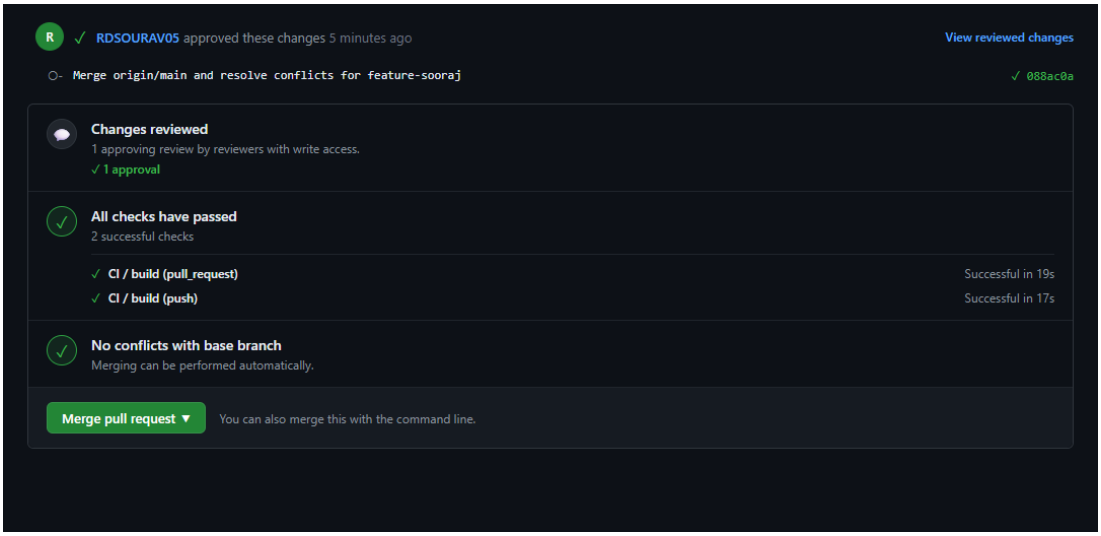


Figure 2 (Screenshot 2 Required): Pull Request #9 approved by teammate, with all checks passed and no merge conflicts.

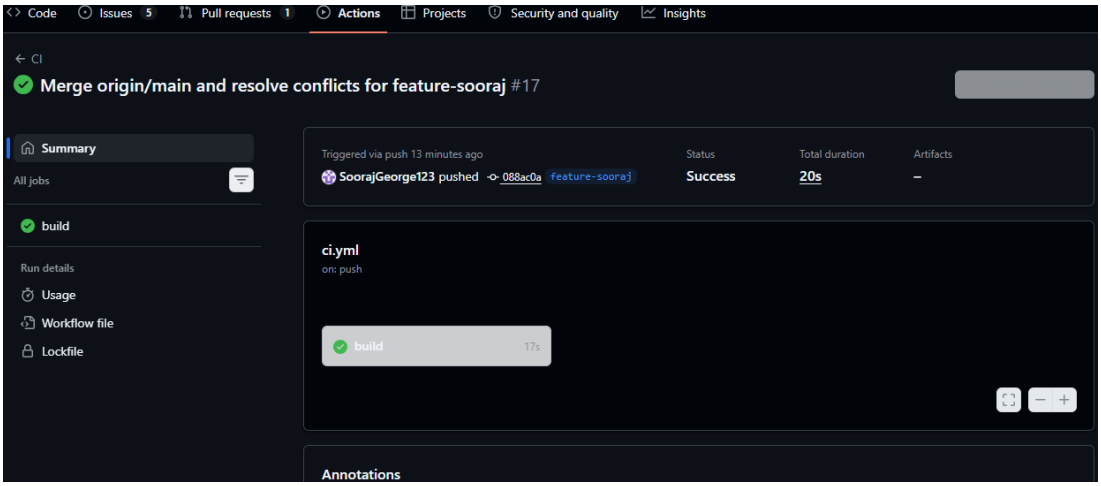


Figure 3 (Screenshot 4 Required): GitHub Actions CI workflow run passing with successful green tick status.

Part 4: CI/CD Status Badge & Agile Project Management

The passing CI/CD status badge was copied and placed at the top of `README.md` displaying real-time build and test health. An Agile Kanban Board ('Experiment 7 Board') was established with three columns: **Todo, In Progress, and Done**. Issue #3 ('Implement testing for Face Detection Module.') was added to the board and tracked actively in the **In Progress** column during module test development.

README

ci.yml passing

SAFE-VISION: AI-Powered Media Moderation & Quality Assurance System

Abstract

SAFE-VISION is an automated multimodal computer vision and quality assurance platform designed for content safety, advertisement moderation, and dataset acquisition. The system processes multimodal ad media and face detection inputs, categorizing content into universal safe categories (SAFE_FOR_ALL) and age-gated categories (SAFE_14_PLUS). This repository manages the core engineering practices, including automated PyTest unit test suites, Git feature-branch workflows, and GitHub Actions CI/CD pipelines.

Team Members & Project Roles

Name	Roll No	GitHub Username	Assigned Module
Sourav	51	RDSOURAV05	Dataset Acquisition & Management

Figure 4 (Screenshot 5 Required): Repository README showing the passing CI status badge and project abstract.

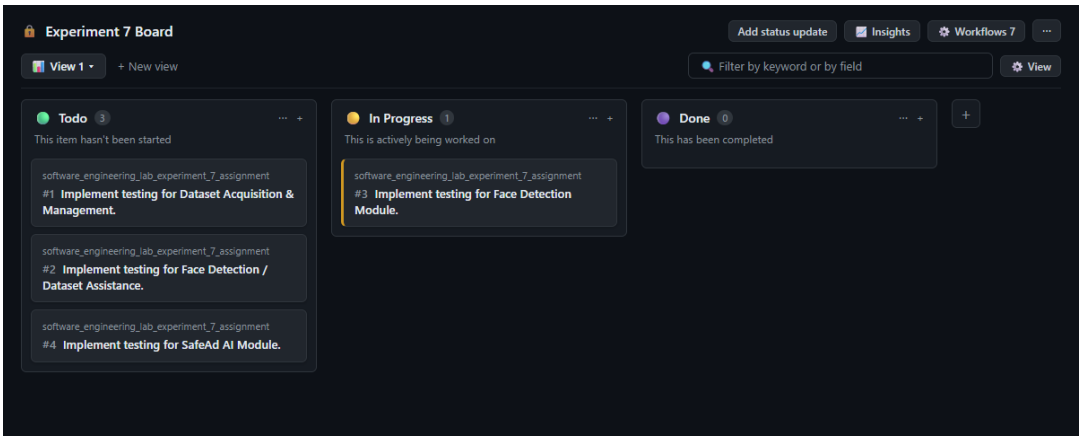
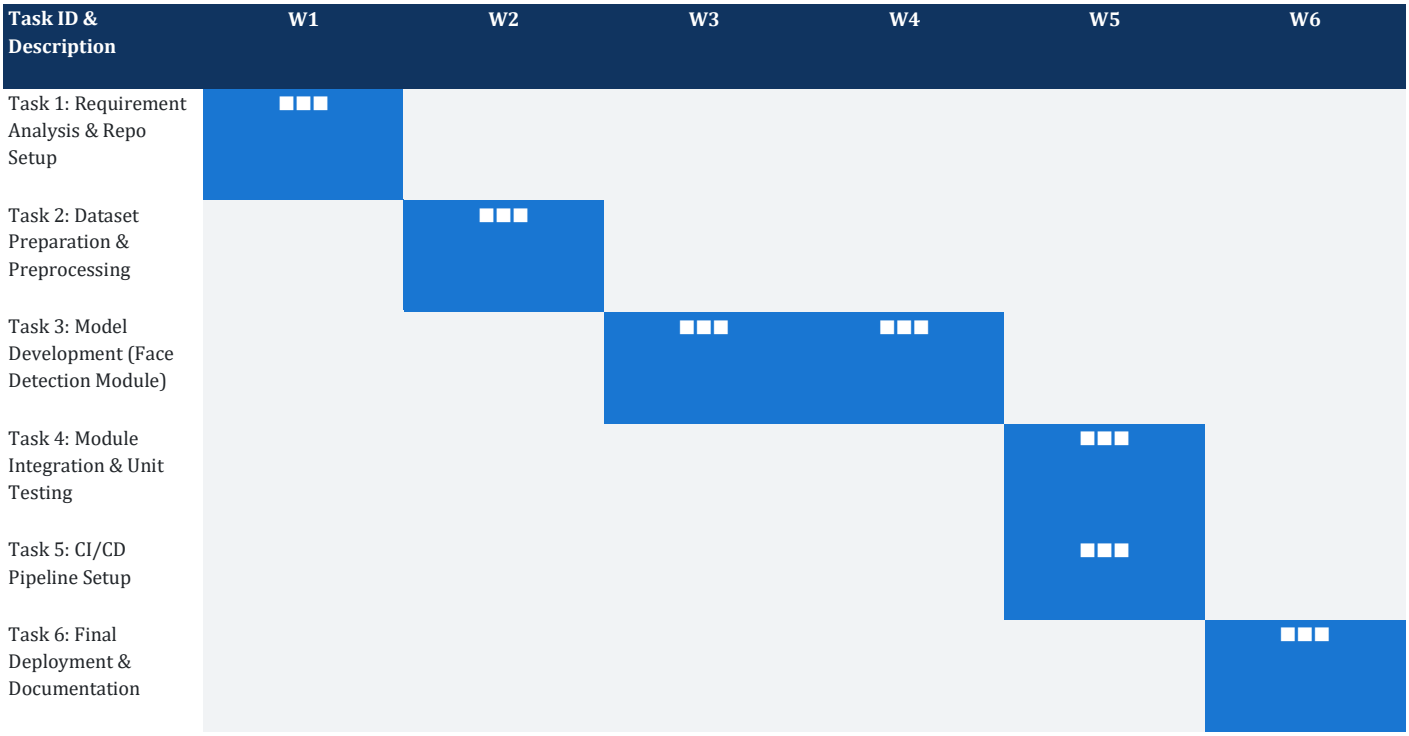


Figure 5 (Screenshot 3 Required): Kanban Board showing Issue #3 in the 'In Progress' column.

Part 5: Gantt Chart & Risk Management

Project Execution Timeline (Gantt Chart - 6 Tasks):



Risk List & Mitigation Strategies (Exact Specification):

1. **Risk 1 (Technical):** Inaccurate AI classification. **Mitigation:** Use a diverse dataset and adjust risk score thresholds.
2. **Risk 2 (Operational):** GitHub merge conflicts. **Mitigation:** Commit frequently and always peer-review pull requests.
3. **Risk 3 (Technical):** CI/CD pipeline failures due to dependencies. **Mitigation:** Pin specific package versions in requirements.txt.

Part 6: Test Case Specification & Verification Table

Unit Test Case Results (Step 4.3 Table):

TC ID	Function Tested	Input	Expected Output	Actual Output	Pass/Fail
TC-01	detect_faces	valid_face.jpg	>= 1	1	**Pass**
TC-02	detect_faces	empty.jpg	ValueError	ValueError	**Pass**
TC-03	detect_faces	corrupt.txt	IOError (EP)	IOError (EP)	**Pass**
TC-04	detect_faces	zero_res.jpg	ValueError (BVA)	ValueError (BVA)	**Pass**
TC-05	detect_faces	video.mp4	ValueError (EP)	ValueError (EP)	**Pass**

(Note: BVA = Boundary Value Analysis, EP = Equivalence Partitioning)

Part 7: Cloud Deployment Architecture & Strategy

Deployment Note (Exact Specification):

"The SAFE-VISION project can be deployed on an Infrastructure as a Service (IaaS) platform such as AWS EC2, or a Platform as a Service (PaaS) such as Heroku. Because the technology stack is Python-based (FastAPI and Streamlit), the application can be seamlessly packaged as a Docker container by configuring a Dockerfile, ensuring a consistent execution environment from local testing to cloud production."

Part 8: Final Submission Verification Checklist

- [✓] Screenshot 1: Assigned Issue (Issue #3 with SoorajGeorge123)
- [✓] Screenshot 2: Approved Pull Request (PR #9 approved by peer)
- [✓] Screenshot 3: Kanban Board (Issue #3 in 'In Progress' column)
- [✓] Screenshot 4: Successful GitHub Actions run (Green Tick status)
- [✓] Screenshot 5: README file showing passing status badge
- [✓] The Gantt Chart (6 Tasks covering project lifecycle)
- [✓] The Risk List (3 Technical & Operational risks with mitigations)
- [✓] The Test Case Table (5 BVA and EP test cases passed)
- [✓] The Deployment Note (Cloud architecture for SAFE-VISION)
- [✓] Submission Format: Exact A2_60_Sooraj.pdf